Al- Balqa' Applied University
Faculty of Engineering
Technology
Physics and basic science
Department

Time: 50 minut
Second Exam Li
algebra
Sunday, Dec, 27,

مدرس المادة: شادي الشوالقة
موعد المحاضرة: ١٠،١١ حد ث، خم

Answer the following five questions:

1- Let X and Y be two vectors in R^n. Prove that

Hint: $|X.Y| \le \|X\|\,\|Y\|$.

2- Verify the triangle inequality for $X = [1\ 3\ 2\ -1]^T, Y = [1\ 0\ -3\ 2]^T$

3- Let $X = x_1 i + x_2 j + x_3 k$, $Y = y_1 i + y_2 j + y_3 k$, $Z = z_1 i + z_2 j + z_3 k$
Show that

4- Find the area of the triangle with vertices X_1, X_2, X_3 where
$X_1 = (1,-2,4)$, $X_2 = (-3,1,4)$, $X_3 = (0,4,3)$.

5- Let V be the set of all continuous function on $(0,\infty)$ and W be the set of all continu
functions on $[1,b]$. Prove that W is subspace of the vector space V
under $f \oplus g = (f+g)(x)$, and $\alpha \odot f = (\alpha f)(x)\ \forall f,g \in V$, *scalar* α

Good Luck

SHADI SHAWAQFEH

Prob. C.5: Answer by True (*T*) or False (*F*):

1 If A is an n×n nonsingular matrix, then the row-echelon form of A is I_n ( )
2 If A is a singular n×n matrix, and B results by interchanging two rows of A, then B is singular ( )
3 If $B = 2A$, then $B^{-1} = 2A^{-1}$ ( )
4. If A is a 4×4 orthogonal matrix, then rank$(A) < 4$ ( )
5 If A and B are n×n matrices, then $(A + B)^2 = A^2 + 2AB + B^2$ ( )
6 If A and B are n×n matrices, then $(A + B)(A - B) = A^2 - B^2$ ( )
7. If A is a 4×4 matrix such that the homogeneous system $AX = 0$ has only the trivial solution, then rank$(A) < 4$. ( )
8. If det$(A) = 2$, then det$(A^{-1}) = 0.5$ ( )
9. If A is an n×n nonsingular matrix, then the columns of A span R^n ( )
10. If the row-echelon form of A has a row of zeros, then there exist nontrivial solutions. ( )
11 If A is an n×n singular matrix, then the system $AX = B$ has exactly one solution for every B ( )
12. If $A^{-1} = A^T$, then A is orthogonal

Prob. C.6: What conditions must b_1, b_2, and b_3 satisfy in order the following system of equations to be consistent?

$$\begin{aligned} x + y + 2z &= b_1 \\ 2x + y + 3z &= b_2 \\ x \qquad + z &= b_3 \end{aligned}$$

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Prob. C.7: Consider the set V of all vectors in R^3 and define the operations by

X⊕Y = X + 2Y **c.X = 3cX**

Determine whether of the following operations are satisfied.

1. X⊕(Y⊕Z) = (X⊕Y)⊕Z
2. X⊕(−X) = 0
3. c.(X⊕Y) = (c.X)⊕(c.Y)
4. (c + d).X = (c.X)⊕(d.X)

1. ....

2. ....

3. ....

4. ....

Prob. C.8: Consider the matrix A = [1 0 0 ; 0 *cos(t)* *sin(t)* ; 0 *−sin(t)* *cos(t)*]

Using factor expansion find the inverse of the matrix A

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#امتحان فاينل

* سؤال بيعطيك مستقيمين وتوجد ال [area of parrallelogram
Volume ,, ,,]

* سؤال بيعطيك مصفوفة وبيطلب

1. The Polynominal
2. The eigenvalues of A; ...
3. The Rank of matrix
4. The inverse of matrix

* سؤال [صح وخطأ] ب 20 علامة

* سؤال حل ع اي موضوع ويكون فاينل أسر

Let $x_1 = (1,1,1)$ $\quad x_2 = (0,1,1)$

$x_3 = (1,0,1)$

$S = (x_1, x_2, x_3)$

#PCE-Fet

1] Is S a basis of R^3? why?

2]

أسئلة سنوات سابقة

المادة: جبر خطي

Q1 - Find Basis For Null space of $A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}$

Q2 - Find Basis For column space of $A = \begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ -1 & -6 & 2 \end{bmatrix}$

Q3 - if $AB = C$ where $B = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & 2 \end{bmatrix}$,

$C = \begin{bmatrix} -5 & -1 & 0 \\ 6 & -3 & 7 \end{bmatrix}$ Find A.

Q4 - $V = \{(x, y, z) / x, y, z \in R\}$, Define
$(x_1, y_1, z_1) \oplus (x_2, y_2, z_2) = (x_1 + x_2, y_1 - y_2, z_1 + z_2)$
give counter example to show that V is not vector space.

5 - $T : P_1 \longrightarrow R^2$ is linear transforms
$T(2t+1) = (-2, 1)$ and $T(t-2) = (3, 2)$
Find $T(5t-10)$.

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